

Kenneth Emata

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SKILLS

- **Languages:** C/C++, Rust, Java, Python, JavaScript, SQL, R, Bash
- **Tools & Platforms:** Git/GitHub, Docker, VS Code, Linux, MATLAB, NeoVim, MS Office, Google

WORK EXPERIENCE

Classroom Facilitator/Curriculum Developer

June 2025 - Present

STEMA Learning Centre, Surrey, BC

- Designed and delivered technical curriculum for 30+ students ages 4–13, translating programming and engineering fundamentals into structured, age-appropriate lessons.
- Built programming modules in Python, JavaScript, HTML/CSS, and Scratch covering core concepts including loops, conditional logic, and variables.
- Engineered hands-on engineering lessons using robotics kits, focusing on mechanical assembly, structural stability, and sensor integration
- Oversaw technical operations including hardware maintenance, software configuration, and classroom equipment logistics across multiple sessions.

SOFTWARE PROJECTS

Cellar: CLI Password Manager

September 2026

Technologies: Python, Typer, Argon2id, AES-256-GCM, Git/GitHub

- Designed and implemented a security-focused CLI application from scratch, applying Argon2id key derivation and AES-256-GCM authenticated encryption to protect user data.
- Prevents data loss on crash or concurrent access by writing to a temp file, syncing to disk, and atomically renaming it into place, with a file lock so two instances can't write to the vault at the same time
- Persisted Argon2id's cost parameters (time, memory, parallelism) per vault, so the Key Derivation Function (KDF)'s difficulty can be tuned later without breaking vaults encrypted under older settings.
- Implemented master-password rotation by deriving a new key from the new password and re-encrypting the full vault under it, rather than just replacing the password check.

Turtle Trips: AI-Powered Travel Planner

October 2025

Technologies: Flask, Python, HTML/CSS, Javascript, Git/Github, Gemini API

- Developed a generative AI planner that converts natural language prompts into comprehensive multi-day itineraries covering dining, lodging, and activities.
- Architected a Flask backend to orchestrate Gemini API prompts and process complex data transformations for seamless itinerary generation.
- Built an interactive mapping interface utilizing Google Maps/Places APIs to visualize waypoints, calculate route distances, and optimize travel paths.
- Streamlined geolocation and venue discovery API integrations, delivering high-fidelity, real-time location data to users.
- Led version control and code review workflows for a 4-person team throughout the hackathon, maintaining a clean GitHub history.

EDUCATION

Bachelor of Science in Computing Science

Expected: December 2027

Simon Fraser University, Burnaby, BC